Detecting cancer cells in Acute Myeloid Leukaemia using an Auto-Encoder

Alice Driessen[1,2,*], Susanne Unger[3], An-phi Nguyen[1,2], Burkhard Becher[3], Maria Rodriguez Martinez[1]

[1]IBM Research Europe | Zürich Lab [2]ETH Zürich [3]University of Zürich *adr@zurich.ibm.com

Introduction

Acute Myeloid Leukaemia (AML) is a cancer in the bone marrow of cells in the myeloid lineage. Marker expression-based identification of tumour cells in bone marrow samples is challenging due to the changing marker expression of healthy cells in development, the heterogeneity and complex clonal composition of the cancer, as well as its complex evolution as the disease progresses. We propose an auto-encoder (AE) trained on healthy bone marrow cells to distinguish the tumour cells from the healthy cells.

Dataset

- Flow cytometry data of 20 patients that underwent immunotherapy.
- Expression of 23 markers at diagnosis, remission, and relapse.
- Remission samples of four patients.

Results

- Auto-encoders trained on remission samples ("healthy cells") in cross-validated manner
- The model with lowest reconstruction error captures the developmental trajectory of healthy bone marrow (fig. 1A)
- Using the latent space of this model we can predict the celtype of non-remission cells (fig. 1B)
- Evaluated models in classifying single cells as blasts on syntetic mixutes with a range of known blasts fraction.
- The best model had an average AUROC of 0.95 and clearly outperformed a random model (fig. 1 C&D)
- Predicting both blast and celltype we can analyse and predict the blast fraction and developmental stage of patients leukaemias (fig. 2)

Outlook

Compare our findings to clinical annotations. Identify suitable markers for targeted therapy and identify phenotype of cells that survive therapy.

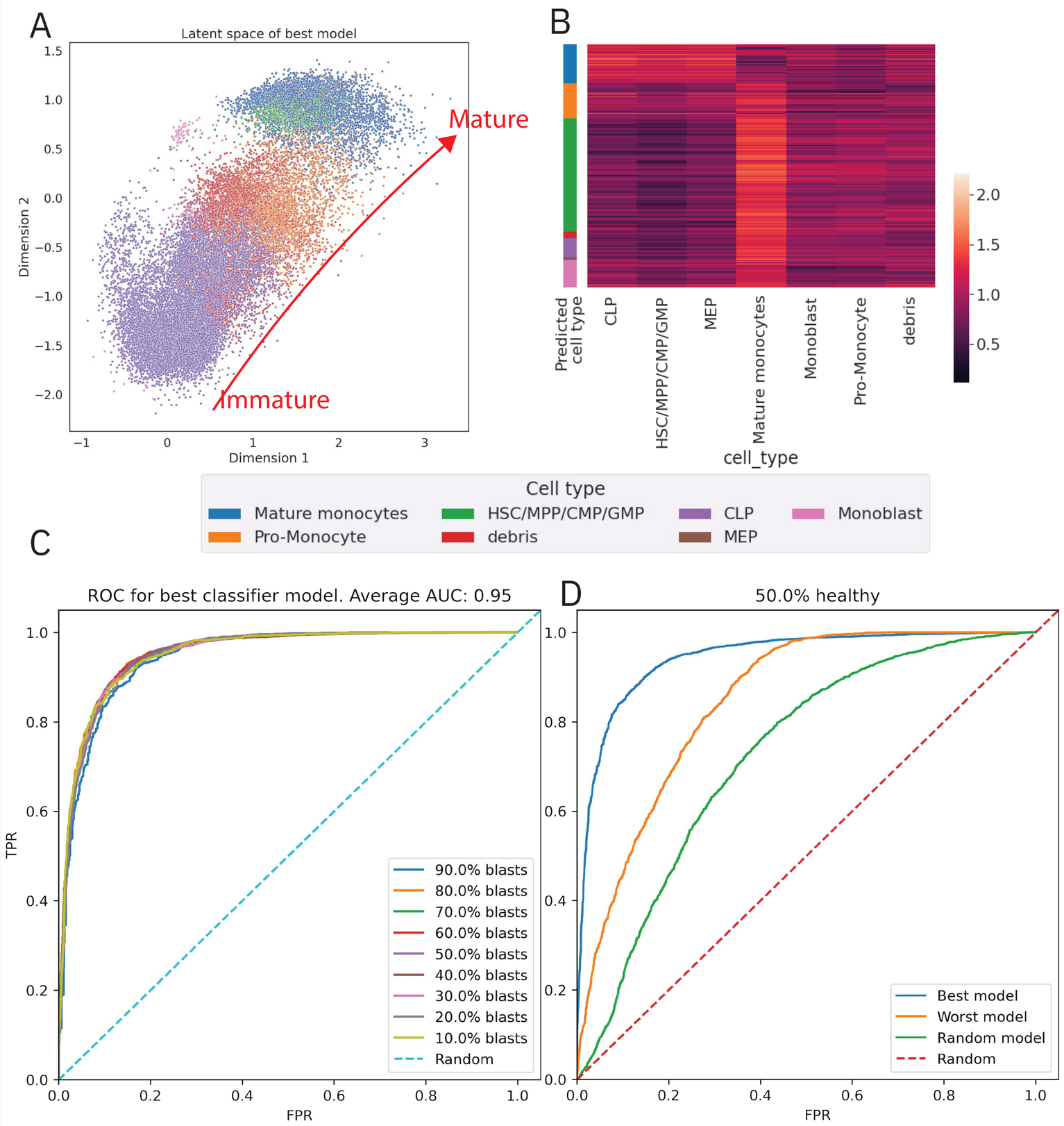

Figure 1. We trained 400 auto-encoders using 100 randomly sampled parameters each trained and validated on 4 different folds of healthy cells. The models were optimized using mean squared reconstruction error (MSE) loss and we selected the top-10 models with lowest MSE. We annotated some non-remission cells with clear abberant expression as blasts and made synthetic samples by mixing these with healthy cells not seen during training. A: Latent space of model with lowest reconstruction error on healthy cells. B: Eucledian distance in full marker space (23) between single cells and the mean expression of the annotated cells. C: Performance of model with highest AUROC on all synthetic samples. D: Model with highest AUROC compared to model with worst reconstruction error and a model with same architecture but random parameters.

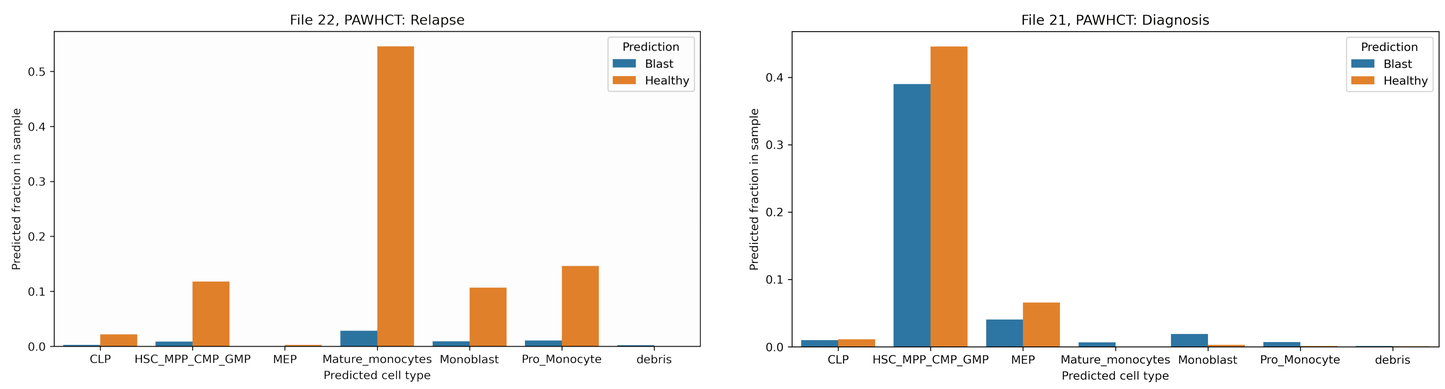

Figure 2 Fraction of cells predicted to be malignant or healthy and predicted to be of a cell type for two samples of one patient. Cell type is predicted with a kNN classifier in the 2dimensional latent space of the model with the lowest MSE on healthy cells. The blast/healthy predction is done with the model with highest AUROC. These samples match the clinical report relatively well.

ETH zürich | University of Zurich[UZH] | IBM Research Europe

We acknowledge funding by the European Union's Horizon 2020 research and innovation programme (iPC–Pediatric Cure, No. 826121).

Individualized PaediatricCure